Experiment ‘p_4_3_e_positions_of_vowels_array’ Results

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Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 2
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 2

Problem Specification

Problem name: p_4_3_e_positions_of_vowels_array
Natural language statement: null
Method signature: p_4_3_e_positions_of_vowels_array(s: string) returns (positions: array<int>)

Ensures

- `forall p :: p in positions[..] ==> 0 <= p < |s| && s[p] in ['a','e','i','o','u','A','E','I','O','U`
- `forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o','u','A','E','I','O','U'] ==> i in positions[..]`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_4_3_e_positions_of_vowels_array(s: string) returns (positions: array<int>)

The method should respect the following contract:

ensures forall p :: p in positions[..] ==> 0 <= p < |s| && s[p] in ['a','e','i','o','u','
    ↪ A','E','I','O','U'], ensures forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o
    ↪ ','u','A','E','I','O','U'] ==> i in positions[..]

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnLwJihORMBWU7lzhooM67NgmmuLA
Model: gpt-5.5-2026-04-23
Created at: 1780654779
Finish reason: stop
Usage: 418 tokens in, and 5552 tokens out

```
{
  var acc: seq<int> := [];
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant forall p :: p in acc ==> 0 <= p < i && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U']
    invariant forall j :: 0 <= j < i && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪ ==> j in acc
  {
    if s[i] in ['a','e','i','o','u','A','E','I','O','U'] {
      acc := acc + [i];
    }
    assert forall p :: p in acc ==> 0 <= p < i + 1 && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U'];
    assert forall j :: 0 <= j < i + 1 && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪  ==> j in acc;
    i := i + 1;
  }
```

```
  positions := new int[|acc|](j => acc[j]);
  assert positions.Length == |acc|;
  assert forall j :: 0 <= j < |acc| ==> positions[j] == acc[j];
  assert forall j :: 0 <= j < |positions[..]| ==> positions[..][j] == acc[j];
  assert positions[..] == acc;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_4_3_e_positions_of_vowels_array/program.dfy(20,38): Error: index out of
    ↪ range
   |
20 | positions := new int[|acc|](j => acc[j]);
   | ^



Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnLyWamrzHEJ92q28iVzhLhZpjmey
Model: gpt-5.5-2026-04-23
Created at: 1780654916
Finish reason: stop
Usage: 878 tokens in, and 2449 tokens out

```
{
  var acc: seq<int> := [];
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant forall p :: p in acc ==> 0 <= p < i && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U']
    invariant forall j :: 0 <= j < i && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪ ==> j in acc
  {
    if s[i] in ['a','e','i','o','u','A','E','I','O','U'] {
      acc := acc + [i];
    }
    assert forall p :: p in acc ==> 0 <= p < i + 1 && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U'];
    assert forall j :: 0 <= j < i + 1 && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪  ==> j in acc;
```

```
    i := i + 1;
  }

  positions := new int[|acc|](j => if 0 <= j < |acc| then acc[j] else 0);
  assert positions.Length == |acc|;
  assert forall j :: 0 <= j < |acc| ==> positions[j] == acc[j];
  assert forall j :: 0 <= j < |positions[..]| ==> positions[..][j] == acc[j];
  assert positions[..] == acc;
}
```

Final Program

```
method p_4_3_e_positions_of_vowels_array(s: string) returns (positions: array<int>)
        ensures forall p :: p in positions[..] ==> 0 <= p < |s| && s[p] in ['a','e','i','o
            ↪ ','u','A','E','I','O','U']
        ensures forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o','u','A','E','I','O','
            ↪ U'] ==> i in positions[..]
{
  var acc: seq<int> := [];
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant forall p :: p in acc ==> 0 <= p < i && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U']
    invariant forall j :: 0 <= j < i && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪ ==> j in acc
  {
    if s[i] in ['a','e','i','o','u','A','E','I','O','U'] {
      acc := acc + [i];
    }
    assert forall p :: p in acc ==> 0 <= p < i + 1 && s[p] in ['a','e','i','o','u','A','E
        ↪ ','I','O','U'];
    assert forall j :: 0 <= j < i + 1 && s[j] in ['a','e','i','o','u','A','E','I','O','U']
        ↪  ==> j in acc;
    i := i + 1;
  }

  positions := new int[|acc|](j => if 0 <= j < |acc| then acc[j] else 0);
  assert positions.Length == |acc|;
  assert forall j :: 0 <= j < |acc| ==> positions[j] == acc[j];
  assert forall j :: 0 <= j < |positions[..]| ==> positions[..][j] == acc[j];
  assert positions[..] == acc;
}
```

Total Token Usage

Input tokens: 1296
Output tokens: 8001

Reasoning tokens: 7161
Sum of 'total tokens': 9297

Experiment Timings

Overall Experiment started at 1780654777109, ended at 1780654992354, lasting 215245ms (215.25 seconds)
Iteration #1 started at 1780654777110, ended at 1780654915953, lasting 138843ms (138.84 seconds)
Iteration #2 started at 1780654915953, ended at 1780654992354, lasting 76401ms (76.40 seconds)